

Epigenomic methods at NGI

Bulk methods across NGI Stockholm and NGI Uppsala —
methylation, chromatin and native base modifications



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One national infrastructure, three nodes

The infrastructure

NGI Stockholm

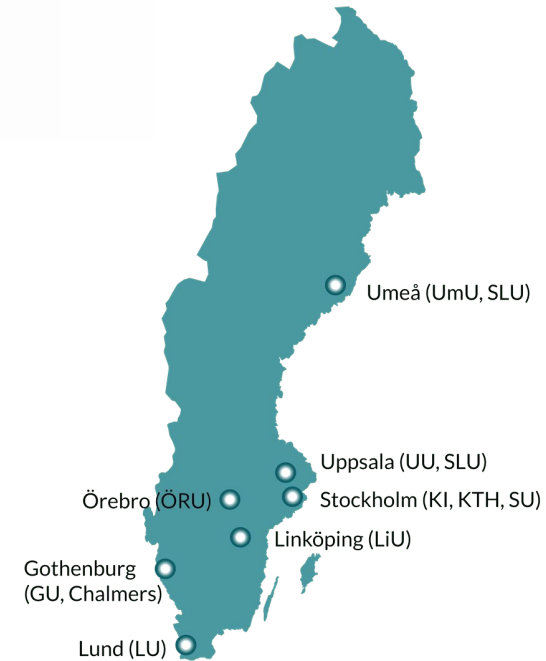
SciLifeLab Solna

Uppsala Genome Center

NGI Uppsala · Uppsala University

SNP&SEQ

NGI Uppsala · Uppsala University



NGI STOCKHOLM IN 2025



740

projects

46,123

samples

1.6 Tbp

sequenced per day, on average

≈ 16 human genomes at 30x

31

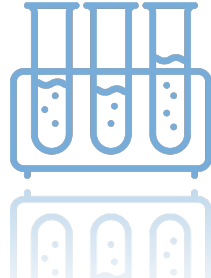
epigenetics projects, 391
samples

From first meeting to data delivery

How a project runs



Meeting



QC & library prep



Sequencing



Analysis & delivery

Common user challenges

- Not knowing enough about their own samples
- No tools for sample quality control
- Limited bioinformatics capacity
- Lost in the literature and the options

What NGI brings

- Experience across very diverse sample types
- Sample QC with clear thresholds and requirements
- Basic data QC, plus a link to NBIS
- One-to-one planning meetings

Eight bulk epigenomic methods across NGI

The menu

Method	What it reads	Site	Status
Illumina EPIC v2 array	5mC at >935,000 human CpGs	NGI Uppsala	Routine
EM-seq	Genome-wide CpG methylation, base resolution	NGI Uppsala	Routine
BSK-SPLAT (WGBS)	Genome-wide CpG methylation, low input	NGI Uppsala	Routine
Omni-ATAC	Open chromatin, nucleosomes, TF footprints	NGI Stockholm	Routine
Dovetail Omni-C	3D contacts, TADs, assembly scaffolding	NGI Stockholm	Routine
EpiFinder ChIP-seq	Up to 8 histone marks, quantitative	NGI Stockholm	Pilot
Nanopore native DNA	Modified bases on long reads	NGI	Routine
PacBio Revio kinetics	m6A and m4C, mainly microbial	NGI Uppsala	Routine

Four ways to read methylation at NGI

DNA methylation

EPIC v2 array

>935,000 CpGs, quantitative
per site

Human only

**Best for: large cohorts,
EWAS (epigenome-wide
association study)**

EM-seq

Whole genome, enzymatic
conversion (TET2 +
APOBEC)
Gentle on DNA, even
coverage

**Best for: methylomes of any
species**

BSK-SPLAT

Whole genome bisulfite,
developed at NGI
Low input, lambda spike-in
control

Developed at NGI
[https://doi.org/10.1093/nar/
gkw1110](https://doi.org/10.1093/nar/gkw1110)

**Best for: WGBS from limited
DNA**

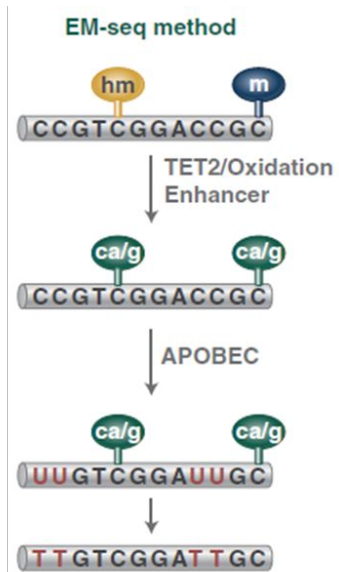
Long-read native

Nanopore or PacBio, no
conversion step
Methylation with haplotypes
and SVs

Best for: phasing, repeats

Four ways to read methylation at NGI

DNA methylation



EM-seq

Whole genome, enzymatic conversion (TET2 + APOBEC)
Gentle on DNA, even coverage

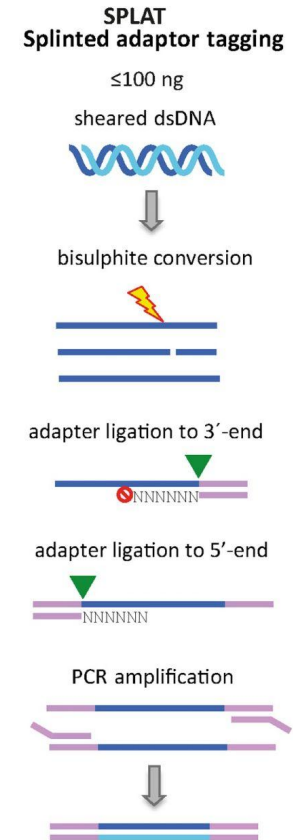
Best for: methylomes of any species

BSK-SPLAT

Whole genome bisulfite, developed at NGI
Low input, lambda spike-in control

Developed at NGI
<https://doi.org/10.1093/nar/gkw1110>

Best for: WGBS from limited DNA

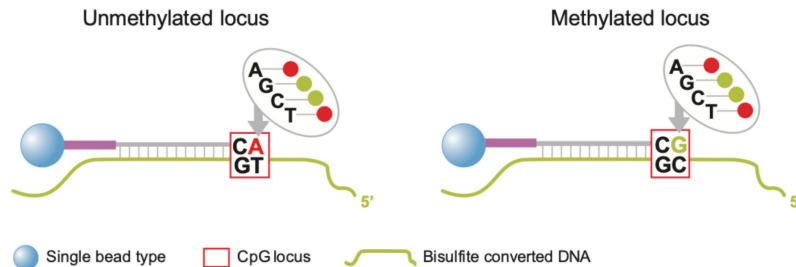


Infinium MethylationEPIC v2.0

DNA methylation · array

Quantitative methylation at single-CpG resolution for human samples. Routine service at NGI Uppsala, ordered through the NGI Order Portal.

Check the sample submission guidelines for input and plate layout before sending.



935k+
CpG sites per sample

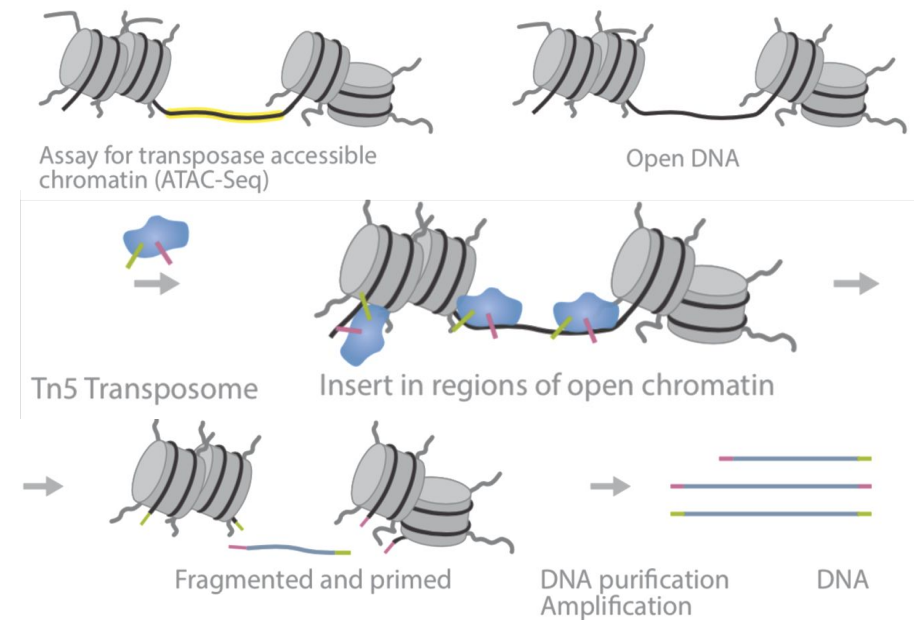
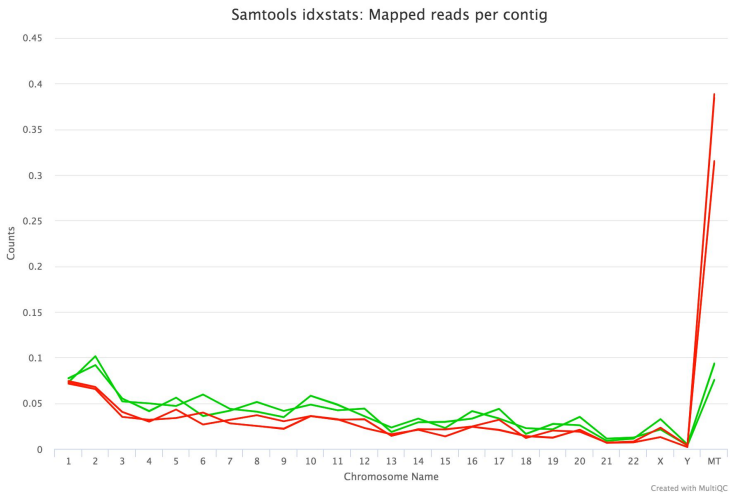
Omni-ATAC

Chromatin accessibility · NGI Stockholm

Maps open chromatin, nucleosome positions and TF footprints; finds candidate enhancers for comparative and evolutionary studies.

Versus standard ATAC-seq

More unique reads and fewer mitochondrial reads, with comparable TSS enrichment (NGI test data)



Requirement	At NGI
Input	50,000 cells or nuclei per tube (~3 Gb genome)
Sample	Fresh cells (>90% viable, Solna only), frozen nuclei, or user-tagmented DNA
Sequencing	>50 M paired-end reads (human)
Batch	Up to 8 samples per day
Analysis	nf-core/atacseq: human, mouse, zebrafish, fly
Library prep	2,639 SEK per sample (academic)

Case from the lab · Omni-ATAC

Over-tagmented PBMC nuclei: a 10 bp sawtooth instead of nucleosomes

At the bench

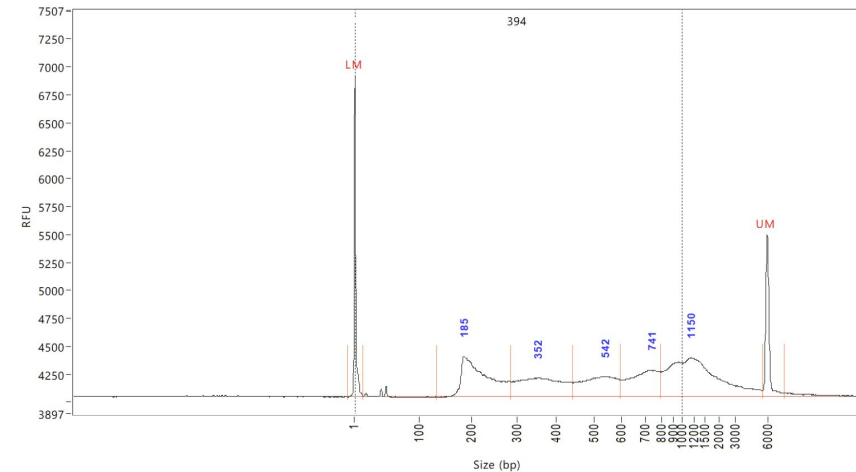
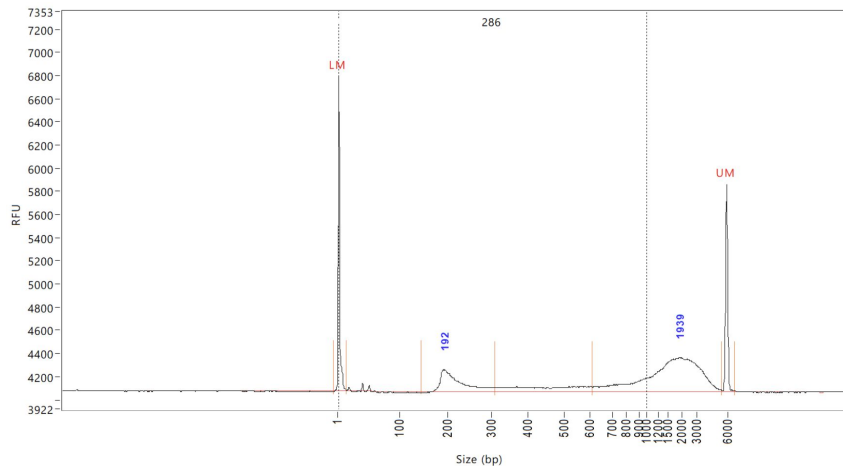
Human PBMC nuclei, tagmented by the user. NGI could only run QC on the finished libraries.

In the QC and data

No nucleosomal wave on the Fragment Analyzer. Insert sizes show a ~10 bp sawtooth starting near 38 bp.

Explanation and fix

Tn5 covers ~38 bp and cuts once per helical turn: it reached free DNA, not chromatin. Options: sequence and assess, or replace samples.



Case from the lab · Omni-ATAC

Over-tagmented PBMC nuclei: a 10 bp sawtooth instead of nucleosomes

At the bench

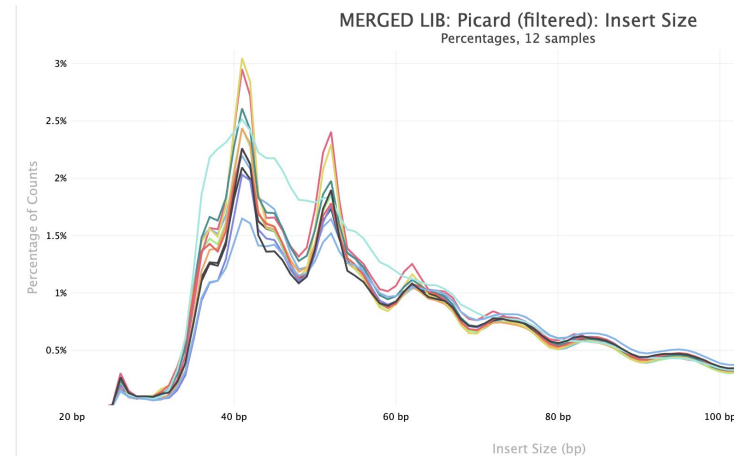
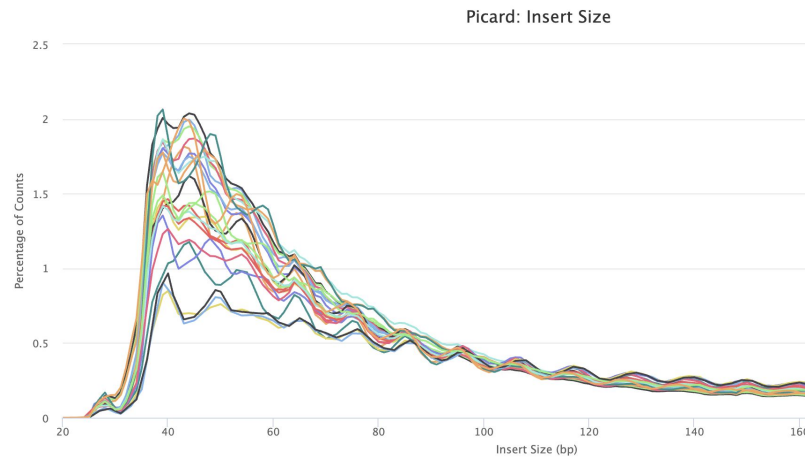
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FOR YOU

Open the insert-size histogram first. Nucleosome peaks in ~200 bp steps are good; a 10 bp pattern alone means naked DNA, more background and unreliable footprints.

Case from the lab · Omni-ATAC

Under-tagmentation tracked the treatment group

At the bench

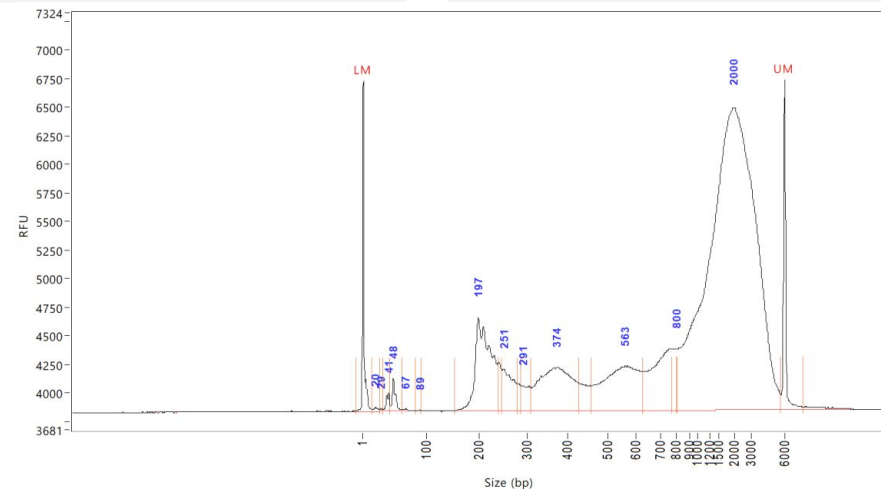
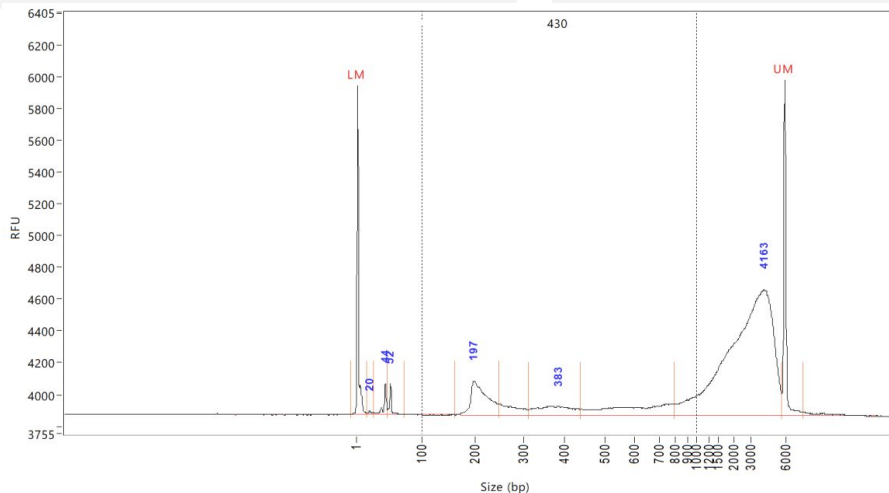
Mouse breast-cancer cell line, knockdown with or without methionine-restricted medium. Nuclei tagmented by the user.

In the QC and data

Every library had a nucleosomal ladder plus a ~2 kb un-fragmented peak, strongest in the methionine-restricted replicates.

Explanation and fix

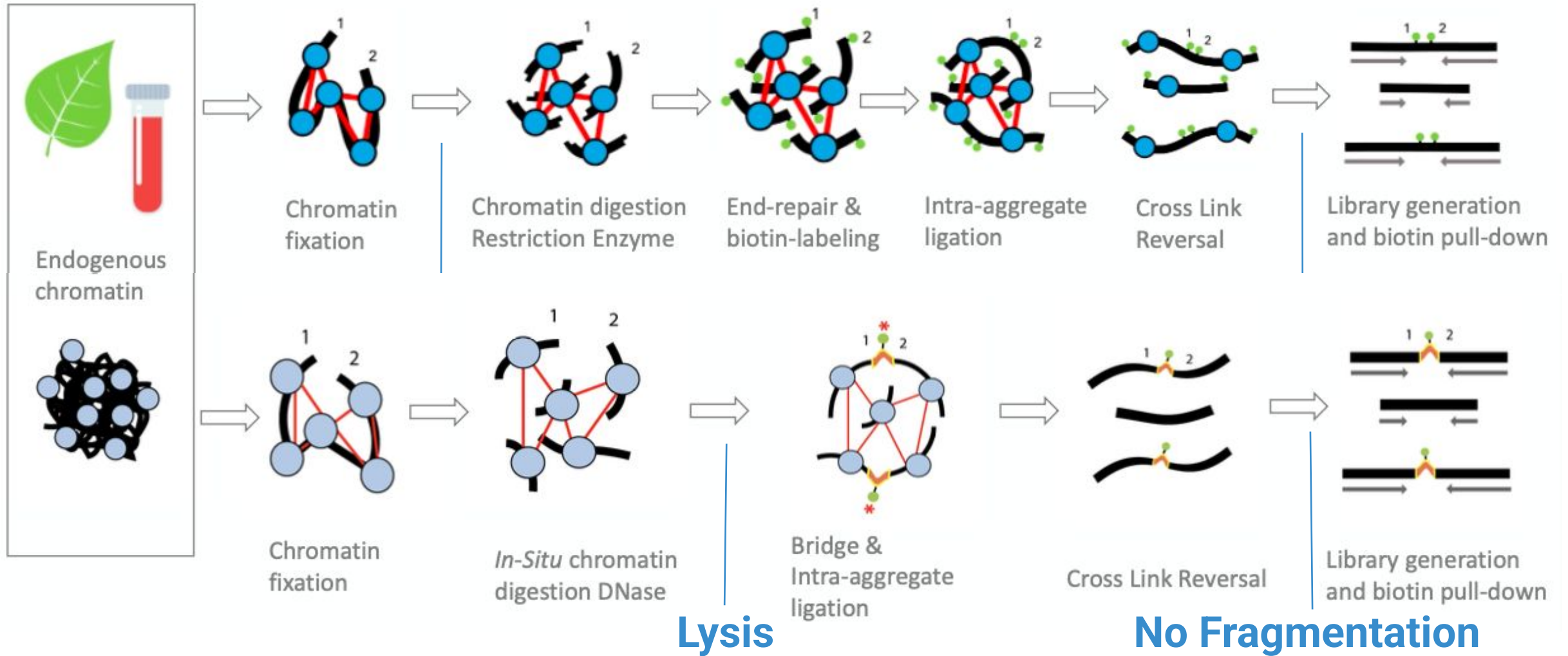
Too many cells or cell-free DNA lower the Tn5:chromatin ratio; compact chromatin was the biological alternative. Sequenced deeper: 50–140 M read pairs.



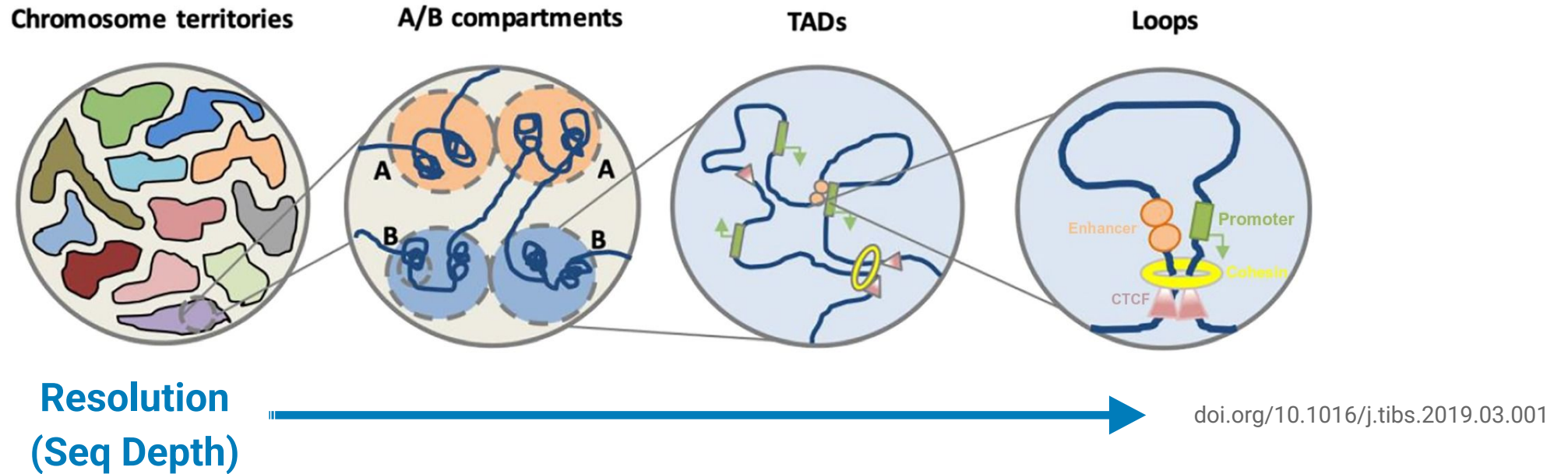
FOR YOU

A technical variable that follows the condition can pass as biology. Compare fragment sizes, TSS enrichment and FRiP per group before calling differential accessibility.

HiC: Restriction Enzyme



Dovetail Omni-C



3D genome · NGI Stockholm

Hi-C with a sequence-independent nuclease instead of a restriction enzyme: even fragmentation, less bias, no DNA extraction.

TADs and loops

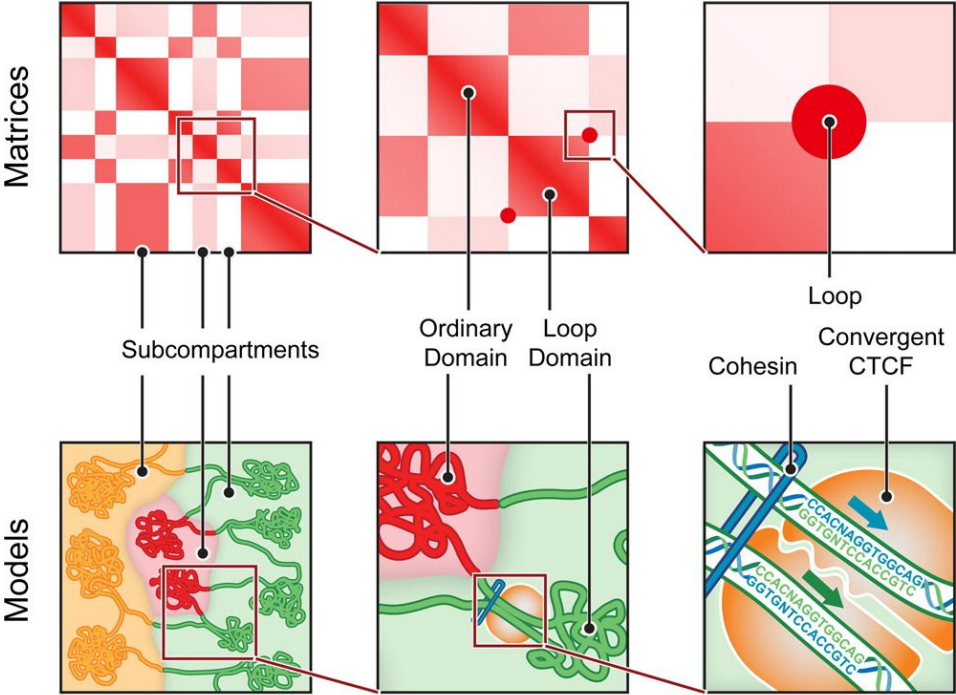
Genome scaffolding

Phasing and SNPs

Analysis included

YaHS draft scaffolding; nf-core/hic for TADs and differential contacts

Dovetail Omni-C



Resolution (Seq Depth) 

doi.org/10.1016/j.cell.2014.11.021

Sample	Input
Animal tissue	20–40 mg soft tissue; ≥50 mg muscle
Blood	1 mL preserved or 3 mL frozen; 10 µL if nucleated
Cells	1 M pelleted (works down to ~200k)
Plant	300 mg young leaves
Depth	~200–300 M reads per Gb genome to start
Library prep	7,217 SEK per sample (academic)
Extra library	5,551 SEK

Case from the lab · Omni-C

Degraded beetle samples gave mostly PCR duplicates

At the bench

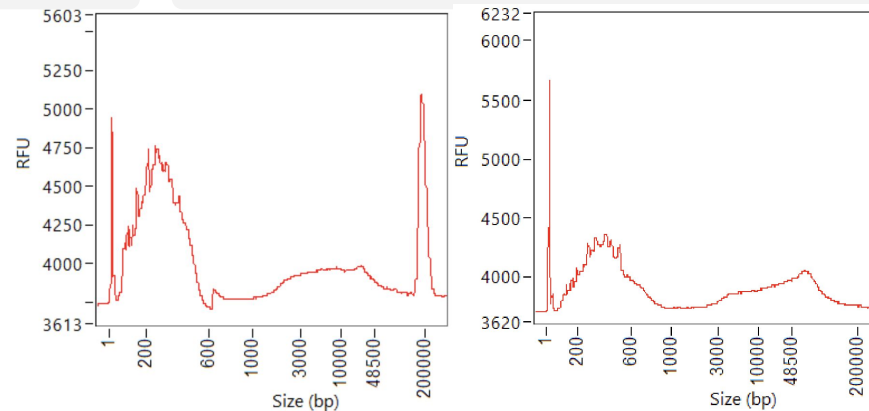
Two beetle species, Omni-C to scaffold HiFi assemblies. The pre-digestion control showed degraded chromatin.

In the QC and data

Lysate hard to quantify, ligation likely overloaded, low library yield. One species scaffolded well; the other was mostly duplicates.

Explanation and fix

Raising PCR from 12 to 16–18 cycles was requested. Extra cycles copy the same molecules: more DNA, no new contacts.



FOR YOU

Library complexity is fixed before PCR. Check the duplicate rate and valid pairs on a shallow run before sequencing deep.

Case from the lab · Omni-C

Isolated nuclei beat ground tissue with half the reads

Perennial flax	Nuclei	Ground tissue
Reads sequenced	522 M	915 M
Duplicates	23.7%	40.0%
Unmapped or one-sided	57.3%	65.6%
Unique valid pairs	16.6%	3.0%
Short cis pairs (<1 kb)	42.9%	28.6%
Informative pairs	79 M	27 M

What we did

One species, two preps. Reads mapped with bwa mem, filtered with pairtools. Informative = cis >1 kb plus trans. Nuclei gave the better scaffolds.

FOR YOU

Judge Hi-C by informative pairs, not total reads or duplicates alone.
For plants: isolated nuclei first, grinding as a last resort.

EpiFinder multiplexed ChIP-seq

Histone marks · NGI Stockholm · pilot

NGI is the first core lab for Epigenica's EpiFinder platform: barcoded, pooled ChIP that gives quantitative tracks across many samples at once.

GenomePro

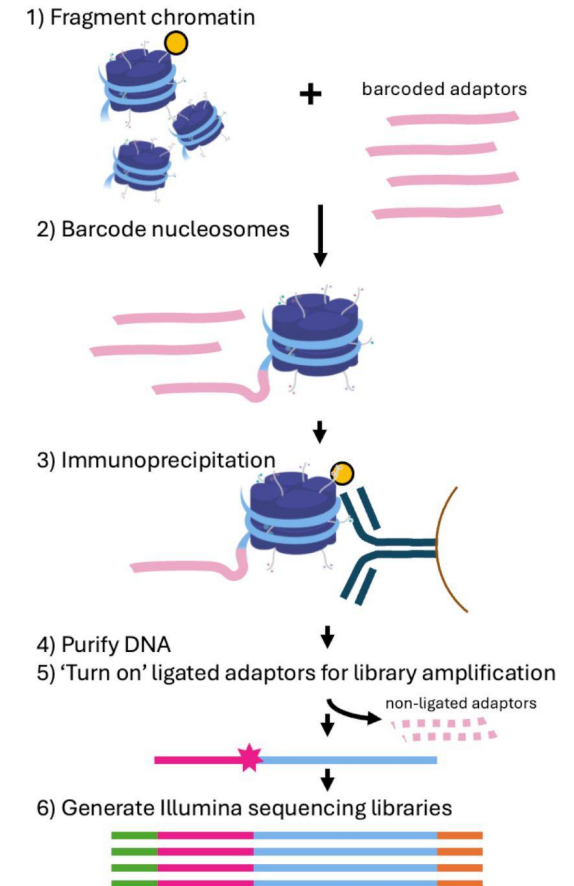
Cells or tissue, frozen or fixed · ~1 M cells

Up to 8 marks x 24 samples = up to 192 quantitative ChIP-seq tracks

cNuc

Plasma or serum · 200 µL

Cell-free nucleosomes: up to 5 PTMs (H3K4me3, H3K27ac, H3K27me3, H3K9me3) plus DNA methylation



Case from the lab · EpiFinder cNuc

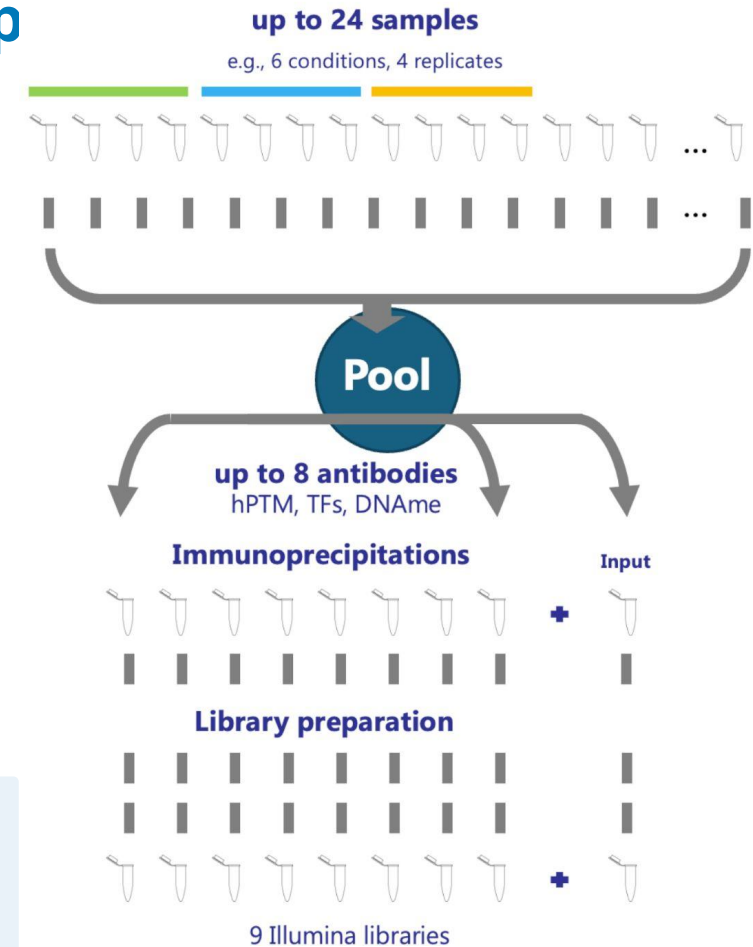
EpiFinder pilots: protocol timing and pooled design shap

Pooled design

24 human serum samples barcoded, then pooled per antibody: 5 marks + 5mC + input = 7 libraries. IP DNA can be below Qubit detection.

Data and depth

FASTQ per antibody, not per sample: demultiplex by internal barcode. Reads per sample: 10 M broad marks, 2.5 M sharp, 5 M input and 5mC.



FOR YOU

Samples in one pool shared the same IP and washes, so compare within a pool. Higher background can come from bead and wash steps, not biology.

Long reads see modifications directly

Native base modifications

No conversion, no amplification: the modification is read from the native molecule, together with haplotypes, repeats and structural variants.

Oxford Nanopore

Modified bases called from the raw current signal, e.g. CpG 5mC

HMW DNA >30 kb · 100–300 fmol (1–3 µg)

Library prep: 2,307 SEK per sample, or 1,194 SEK per sample + 1,372 SEK per pool when barcoded

PacBio Revio

m6A and m4C from polymerase kinetics

Needs deep coverage: mainly microbial genomes

5-10 µg HMW genomic DNA

Nanopore adaptive sampling

Targeted native methylation

Each read is basecalled and aligned in real time; off-target molecules are ejected. No special library prep — just a reference FASTA and a .bed of regions.

- Targets 0.1–10% of the genome ($\leq 5\%$ recommended)
- Add ~10 kb flanks per region
- Typical enrichment 4–6x



bioRx



NGI ExAMPLE RUN

~150x
over a 2 Mb region of interest

20–40x
across the rest of the genome

Case from the lab · Nanopore

Low input gave uneven output and missed the 30x target

At the bench

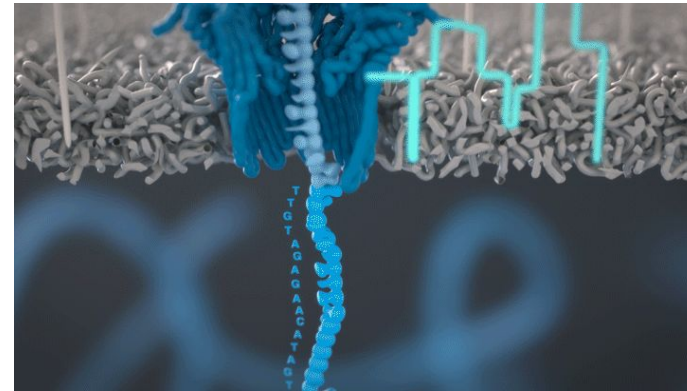
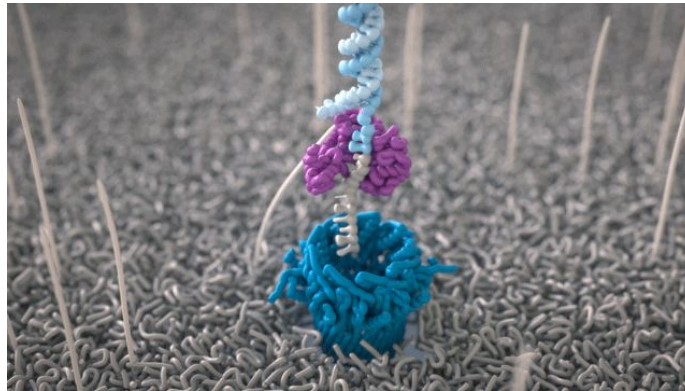
Tissue DNA, adaptive sampling with 5mC and 5hmC calling. 400 ng per sample gave low library yield, so all samples shared one flow cell.

In the data

Output per barcode ranged 1.4–10.8 Gb; regions of interest stayed below 30x. The weakest samples were not those with least DNA.

Explanation and fix

Yield also depends on fragment length, DNA damage and ligation. Re-run: ≥ 1 μ g input, fewer samples per pool, pooling by molarity.



FOR YOU

A methylation frequency is a proportion of reads per site: at 10x, one read moves it by 10%. Carry coverage with every call.

Case from the lab · Nanopore

The extraction kit decided read length for adaptive sampling

At the bench

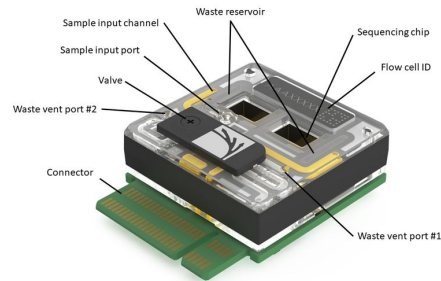
Animal tissue for adaptive sampling.
DNA from a DNA/RNA co-extraction kit with mechanical homogenisation, or an enzymatic HMW kit.

In the QC

TapeStation peaks: 6.6-8 kb after mechanical homogenisation versus 14 kb to over 60 kb after enzymatic lysis.

Explanation and fix

Mechanical homogenisation shears DNA. Only the long-fragment samples were recommended for sequencing.



FOR YOU

Keep-or-eject is decided after a few hundred bases, so short molecules escape enrichment. Check read lengths per sample before comparing enrichment or methylation.

Start from the biological question

Choosing

If you want to know...	Start with
Methylation across a large human cohort	EPIC v2 array
Base-resolution methylome, any species	EM-seq or SPLAT WGBS
Methylation with haplotypes, repeats or SVs	Nanopore ($\pm$ adaptive sampling)
Bacterial methylomes (m6A, m4C)	PacBio Revio
Which regions are open or TF-bound	Omni-ATAC
How the genome folds; assembly scaffolds	Omni-C
Histone marks across many samples	EpiFinder (pilot)

Takeaway

Ask us - even if a method isn't listed, we might run it

Match the protocol

to your biological question → Make sure there is a biological question!

Plan early

sample handling decides success

Know your data

NBIS can help with analysis

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nbis.se